
Machine Learning Model Compression and Optimization for Vehicle Fuel Efficiency Prediction

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Abstract

This project investigates machine learning and neural network approaches for vehicle fuel efficiency prediction using two datasets: the Auto MPG dataset and the Vehicles Fuel Efficiency dataset. Multiple regression models including Linear Regression, Support Vector Regression (SVR), Random Forest Regression, Multilayer Perceptron (MLP), PCA-based MLP, Coreset MLP, and Pruned MLP were implemented and evaluated. For the D3 milestone, an additional PCA-KNN model was introduced as an improvement for the Vehicles dataset. Experimental results show that Random Forest achieved the highest predictive accuracy across both datasets, while PCA significantly improved learning performance on high-dimensional vehicle data. The proposed PCA-KNN pipeline achieved competitive performance, demonstrating the effectiveness of combining dimensionality reduction with instance-based learning.

1 Introduction

Fuel efficiency prediction is an important application of machine learning in transportation, environmental sustainability, and automotive engineering. Accurate predictive models can assist manufacturers in improving vehicle designs and help consumers make informed purchasing decisions.

This project evaluates multiple machine learning and neural network techniques for fuel efficiency prediction. In addition to prediction accuracy, the project investigates model compression and optimization techniques such as Principal Component Analysis (PCA), coreset selection, and pruning. The final phase of the project introduces a PCA-KNN pipeline to improve performance on high-dimensional vehicle data.

2 Datasets

2.1 Auto MPG Dataset

The Auto MPG dataset contains approximately 398 vehicle records with attributes describing engine and vehicle characteristics. The target variable is Miles Per Gallon (MPG). Features include:

- Cylinders
- Displacement
- Horsepower
- Weight
- Acceleration
- Model Year

- Origin

The dataset contains only a small number of missing values and represents a relatively low-dimensional regression problem.

2.2 Vehicles Fuel Efficiency Dataset

The Vehicles Fuel Efficiency dataset contains 47,702 vehicle records and 84 attributes. The target variable used in this study is Combined Fuel Efficiency (comb08).

The dataset contains numerical, categorical, and boolean features as well as substantial missing values. Data preprocessing included missing value handling, feature encoding, feature scaling, and dimensionality reduction.

3 Methodology

3.1 Machine Learning Models

The following machine learning models were evaluated:

1. Linear Regression
2. Support Vector Regression (SVR)
3. Random Forest Regression

3.2 Neural Network Models

The following neural network models were implemented:

1. Standard MLP
2. PCA-Based MLP
3. Coreset MLP
4. Pruned MLP

The standard MLP used three hidden layers with a topology of (64, 32, 16) and was trained using the Adam optimizer.

3.3 Compression and Optimization Techniques

The following optimization techniques were explored:

- Principal Component Analysis (PCA)
- Random Coreset Selection
- Weight Pruning

PCA was used to reduce dimensionality and remove redundant information before model training.

3.4 Proposed Improvement: PCA-KNN Pipeline

For the D3 milestone, a K-Nearest Neighbors (KNN) regressor was introduced for the Vehicles dataset. The execution pipeline consisted of:

StandardScaler → PCA → KNN

First, features were standardized using `StandardScaler`. PCA was then applied to reduce dimensionality and remove noisy features. Finally, a KNN regressor was trained using the transformed feature space. The motivation behind this improvement was that KNN relies heavily on distance calculations, making both scaling and dimensionality reduction essential for effective learning.

4 Results

4.1 Auto MPG Dataset

Table 1 presents the performance metrics of the evaluated models on the Auto MPG dataset. Random Forest achieved the highest predictive accuracy with an R^2 score of 0.913. The standard MLP also performed strongly, while PCA slightly reduced performance due to the already low dimensionality of the dataset.

Table 1: Model Performance on Auto MPG Dataset

Model	MSE	R^2
Linear Regression	8.196	0.848
SVR	6.814	0.873
MLP	5.122	0.905
Random Forest	4.673	0.913
PCA + MLP	6.150	0.886
Coreset MLP	7.268	0.865
Pruned MLP	5.166	0.904

4.2 Vehicles Dataset

Unlike the Auto MPG dataset, the Vehicles dataset contains many more features and records. As shown in Table 2, Random Forest again achieved the highest predictive performance. PCA dramatically improved neural network performance, increasing the R^2 score from 0.052 to 0.863.

Table 2: Model Performance on Vehicles Dataset

Model	MSE	R^2
Linear Regression	97.963	0.200
SVR	121.856	0.005
MLP	116.083	0.052
Random Forest	15.111	0.877
PCA + MLP	16.786	0.863
Coreset MLP	108.010	0.118
Pruned MLP	118.766	0.031

4.3 Improvement Results

The proposed PCA-KNN pipeline achieved competitive performance on the Vehicles dataset, as detailed in Table 3. Although Random Forest remained the best-performing model, PCA-KNN significantly outperformed the original MLP and achieved performance comparable to PCA-MLP.

Table 3: D3 Milestone Improvement Results on Vehicles Dataset

Model	MSE	R^2
PCA + KNN	18.481	0.849

5 Visualizations

The following visualizations were generated during experimentation to evaluate model dynamics and data distributions:

1. PCA t-SNE Projection (Vehicles Dataset)
2. PCA t-SNE Projection (Auto MPG Dataset)

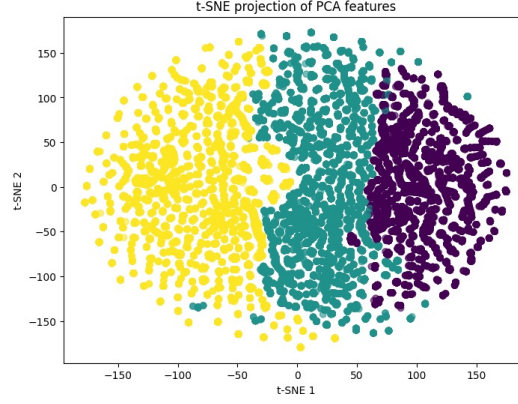


Figure 1: figure
PCA t-SNE Projection for the Vehicles Dataset

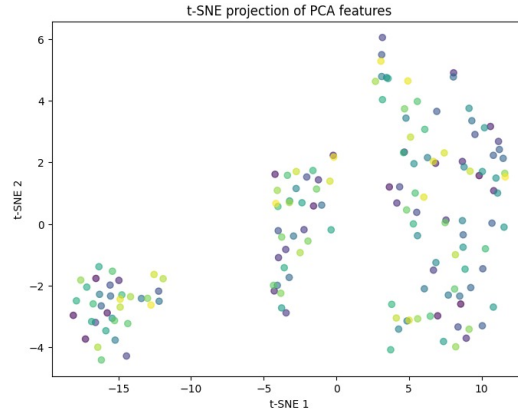


Figure 2: figure
PCA t-SNE Projection for the Auto MPG Dataset

The t-SNE visualizations demonstrate the effectiveness of PCA in separating the data into meaningful clusters and preserving important structural information after dimensionality reduction.

6 Comparison with Existing Studies

The findings of this project align with previous research on fuel efficiency prediction. Siraj et al. demonstrated that neural network models can effectively predict fuel efficiency using the Auto MPG dataset. Sarkar and Hasan (2025) reported strong performance from ensemble learning approaches, particularly Random Forest models. Çeken and Gemci found that machine learning and deep learning methods perform well on Auto MPG and EPA vehicle datasets, with ensemble approaches achieving particularly strong results. The current study supports these findings, as Random Forest consistently achieved the highest predictive performance across both datasets.

7 Discussion

The experimental results demonstrate that dataset characteristics strongly influence model performance. The Auto MPG dataset is relatively small and low-dimensional, allowing most models to perform well. Random Forest achieved the best results, while neural network compression techniques had only minor effects on performance.

The Vehicles dataset presented a more challenging problem due to its size and high dimensionality. PCA proved particularly effective, dramatically improving neural network performance by removing redundant and noisy features. The proposed PCA-KNN pipeline further demonstrated that dimensionality reduction combined with distance-based learning can achieve strong predictive performance. While Random Forest remained the best overall model, PCA-KNN provided a simpler alternative with competitive accuracy.

8 Conclusion

This project compared machine learning, neural network, and model compression techniques for vehicle fuel efficiency prediction using two datasets. Random Forest consistently achieved the highest predictive accuracy across both datasets. PCA was highly effective for improving neural network performance on the high-dimensional Vehicles dataset, while pruning reduced model complexity with minimal performance degradation.

As an improvement introduced during the D3 milestone, a PCA-KNN pipeline was developed and evaluated. The model achieved an R^2 score of 0.849 on the Vehicles dataset, demonstrating that dimensionality reduction combined with instance-based learning provides a viable alternative to neural network-based approaches for fuel efficiency prediction.